Assignment-2: Predictive Modeling Using Linear Regression

Objective

The goal of this assignment is to apply the complete machine learning workflow/pipeline using Linear Regression. You will define a real-world predictive problem, preprocess a raw dataset, build and evaluate a regression model, and implement advanced improvements like hyperparameter tuning and regularization.

Assignment Structure & Requirements

Your final submission must be a structured report containing the following four sections:

- **1. Introduction & Problem Statement**
  • Describe the real-world challenge you are trying to solve. What is the continuous target variable (\(Y\)) you are predicting, and why does this prediction matter to educational institutions? You are required to support your arguments with at least two formal references (e.g., academic studies, institutional research reports) that prove the real-world significance of this problem and validate your solution's potential impact.
- • The Data: Introduce your dataset. Mention the source, and describe at least 5 key features (X) you believe will influence your target variable.
  • The Approach: Explain how you plan to solve this problem. Briefly outline your analytical strategy and why Linear Regression is an appropriate starting baseline.
- **2. Data Preprocessing & Exploratory Data Analysis (EDA)**
  Raw data is rarely ready for machine learning. You must document how you clean and prepare your data:
  • Missing Data & Outliers: Identify any missing values or extreme outliers and justify how you handled them (e.g., imputation, removal).
  • Feature Engineering & Encoding: Convert categorical variables into numerical values using techniques like One-Hot Encoding or Label Encoding.
  • Feature Scaling: Apply scaling techniques such as Standardization (Z-score) or Min-Max Normalization and explain why scaling is crucial for regularization.
  • Data Splitting: Split your dataset into Training (80%) and Testing (20%) sets to ensure unbiased evaluation.
- **3. Baseline Model & Evaluation**
  • **Training**: Fit a standard Ordinary Least Squares (OLS) Linear Regression model on your training data.
  • **Evaluation Metrics**: Evaluate your model on both the training and testing sets. Report and interpret: Mean Absolute Error (MAE), Mean Squared Error (MSE) or Root Mean Squared Error (RMSE).
  • **Overfitting Check**: Compare your training error against your testing error to determine if your baseline model is overfitting or underfitting.

- **4. Model Improvements & Hyperparameter Tuning**
  Linear regression can be improved by adding regularization penalties to prevent overfitting and manage multi-collinearity.
  • **Regularization Techniques**: Implement Ridge Regression (L2 penalty) and Lasso Regression (L1 penalty).
  • **Hyperparameter Tuning**: Use GridSearchCV or RandomizedSearchCV with Cross-Validation (e.g., k=5) to find the optimal regularization strength parameter (alpha or lambda). Explain the technique you pick in terms of how it works, its advantages and disadvantages. Track and log the total wall-clock execution time taken for this search process.
  • Final Comparison: Compare your tuned Ridge/Lasso models against your baseline OLS model using a clear table. Discuss how tuning affected the performance and feature coefficients. Show if it deserves to apply tuning in terms of time.

- Your analysis must include at least three distinct data visualizations (such as graphs or charts) to support your findings and your conclusions

You need to choose of the following project:

1. Student Academic Retention & Graduation GPA Prediction (link)
2. Forecasting E-Commerce Delivery Delay Times (link)

Submission Deliverables

- The **GitHub link** to your code. Your GitHub account should have a readme file that gives a clear description of the problem, how to run the code, etc.
- **Code File:** A fully executable .py script or .ipynb Jupyter Notebook. Ensure your code is clean, commented, and reproducible (use a random_state).
- **Written Report:** A PDF report addressing all the conceptual questions outlined above.